DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : PRE-BOARD EXAMINATION EXAMINATION-II

Subject : Science (086)

SET–II

TIME : 3 Hrs.

M.M. : 80

Date :_____ Name : _____ Class/Sec X_____

General Instructions:

(i) This question paper consists of 39 questions in 3 sections. Section A is Biology; Section B is Chemistry and Section C is Physics.

(ii) All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

Section A

1. Choose the event that does not occur in photosynthesis. (1)
(a) Absorption of light energy by chlorophyll
(b) Reduction of carbon dioxide to carbohydrate
(c) Oxidation of carbon-to-carbon dioxide
(d) Conversion of light energy to chemical energy.

2. The substance that triggers The fall of mature leaves and fruits from plants is: (1)
(a) Auxin
(b) Gibberellins
(c) Abscisic acid
(d) Cytokinin

3. When air is blown from mouth into a test tube containing lime water, the lime water turns milky due to the presence of: (1)
(a) Oxygen
(b) Carbon dioxide
(c) Nitrogen
(d) Water vapour

4. Two pink coloured flowers on crossing resulted in 1 red, 2 pink and 1 white flower progeny. The nature of the cross will be: (1)
(a) Double fertilization
(b) Cross fertilization
(c) Self-pollination
(d) No fertilization

5. In human males, all the chromosomes are paired perfectly except one. This /these unpaired chromosomes is/are: (1)
(i) Large chromosome
(ii) Small chromosome
(iii) Y- chromosome
(iv) X-chromosome.

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(a) (I) and (ii)
(b) (Iii) only
(c) (iii) and (iv)
(d) (ii) and (iv)

6. The maleness of a child is determined by: (1)
(a) The X - chromosome in the zygote.
(b) The Y- chromosome in the zygote.
(c) The cytoplasm of the germ cell which determines the sex.
(d) Sex is determined by chance.

7. Which among the following is **NOT** the function of testis at puberty? (1)
(i) Formation of germ cells.
(ii) Secretion of testosterone.
(iii) Development of placenta.
(iv) Secretion of oestrogen.
(a) (i) and (ii)
(b) (ii) and (iii)
(c) (iii) and (iv)
(d) ((i) and (iv)

The following two questions consist of two statements- assertion (A) and reason(R). Answer the following question by selecting the appropriate options given below.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. **Assertion** (A) : Pepsin acts in the stomach and helps in protein digestion
Reason (R): Pepsin requires an alkaline medium for its activity. (1)

9. **Assertion** (A): In single celled organisms, diffusion is sufficient for transport of substances
Reason (R): The distance of transport of substance within the organism is very small. (1)

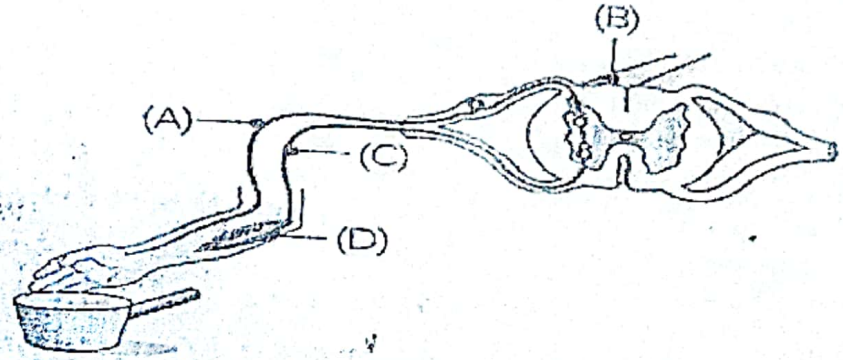
10. DDT was sprayed in a lake to regulate breeding of mosquitoes. How would it affect the trophic levels in a food chain associated with a lake? Justify your answer. (2)

11. Explain the mechanism by which water is transported in plants. (2)

OR

How do the Guard cells regulate opening and closing of stomatal pores? (2)

12. Label the parts A,B,C and D. Show the direction of flow of electrical signals in the given figure. (2)



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13. (i) Why do walls of trachea not collapse when there is less air in it. (1)

(ii) Why is the rate of breathing in aquatic organisms much faster than in terrestrial organisms? (2)

14. Why is damage to the ozone layer a cause for concern? What are its causes and what steps are being taken to limit this damage? (3)

15. Seema crossed pure breed of pea plants having round-yellow seeds with wrinkled green seeds and found that only A – B type of seeds were produced in the F1 generation. When in F1 generation, pea plants having A – B type of seeds were crossbred by self-pollination, then in addition to the original round yellow and wrinkled green seeds, two new varieties A – D and C – B types of seeds were also obtained.

Attempt either subpart (a) or (b)

(a) What are A – D and C – B type of seeds? (2)

OR

(b) Which one will be produced in minimum and maximum number in the F2 generation? (2)

(i) A – B

(ii) C – D

(c) What are A – B type of seeds? (1)

(d) State whether A and B are dominant traits or recessive traits. (1)

16. **Attempt either option A or B**

A. (i) Draw a diagram showing germination of pollen on stigma of a flower and mark on it the following organs/parts. (1+1/2×4=3)

(a) Pollen grain

(b) Pollen tube

(c) Stigma

(d) Female germ cell

(ii) State the significance of pollen tube. (1)

(iii) Name the parts of a flower that develop after fertilization into (i) seed (ii) fruit. (1)

OR

B. (i) Describe the role of prostate gland, seminal vesicle and testes in the human male reproductive system. (3)

(ii) How is the surgical removal of unwanted pregnancy misused? (1)

(iii) Explain the role of oral contraceptive pills in preventing contraception. (1)

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SECTION -B

17. Which of the following reactions is an endothermic reaction? (1)
(a) Burning of coal
(b) Decomposition of vegetable matter into compost
(c) process of respiration
(d) Decomposition of calcium carbonate into quick lime and carbon dioxide.

18. If a few drops of concentrated acid accidentally spills over the hand of a student, what should be done? (1)
(a) Wash the hand with saline solution.
(b) Wash the hand immediately with plenty of water and apply a paste of sodium hydrogen carbonate
(c) After washing with plenty of water apply solution of sodium hydroxide on the hand
(d) Neutralize the acid with a strong alkali.

19. An element X reacts with oxygen to give a compound with a high melting point. The compound is also soluble in water. The element X is likely to be (1)
(a) Iron
(b) Calcium
(c) Carbon
(d) Silicon

20. $CH_3\text{-}CH_2\text{-}OH \longrightarrow CH_3\text{-}COOH$
In the above reaction, alkaline potassium permanganate ($KMnO_4$) acts as: (1)
(a) Reducing agent
(b) Catalyst
(c) Dehydrating agent
(d) Oxidizing agent

21. What happens when dilute hydrochloric acid is added to iron fillings? Choose the correct answer. (1)
(a) Hydrogen gas and iron chloride are produced
(b) Chlorine gas and iron hydroxide are produced
(c) No reaction take place
(d) Iron salt and water are produced

22. A Visually challenged student has to perform a lab test to detect the presence of acid in a given solution. The acid base indicator preferred by him will be: (1)
(a) Blue litmus
(b) Clove oil
(c) Red cabbage extract
(d) Hibiscus extracts

23. Which of the following salts does not contain water of crystallization? (1)
(a) Blue vitrol
(b) Washing soda
(c) Baking soda
(d) Gypsum

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24. Assertion (A): Aluminium is more reactive than iron yet it does not corrode easily. (1)
Reason (R): Aluminium forms a protective oxide layer that prevents further reaction.

25. How the following substances will dissociate to produce ions in their solutions? (2)
(i) Hydrochloric acid
(ii) Nitric acid
(iii) Sulphuric acid
(iv) Magnesium hydroxide.

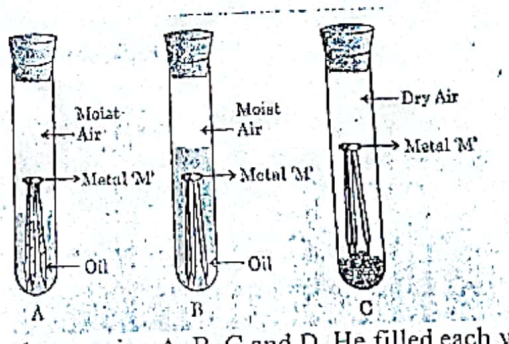
26. One gram of copper powder was taken in a China dish and heated. What change take place on heating? When hydrogen gas is passed over this heated substance a visible change is seen in it. Give the chemical equations of reactions, the name and the colour of the products formed in each case. (3)

27. **Attempt either option A or B**

A. (a) Explain the formation of calcium chloride with the help of electron dot structure. (Atomic number of calcium =20 and chlorine =17). (2)
(b) Why do Ionic compounds not conduct electricity in solid state but conduct electricity in molten and aqueous state? (1)

OR

B. Observe the following diagram showing an experiment to determine the conditions under which a metal M corrodes
List your observations in each of the three cases A, B and C with reason, if the metal M is generally protected against corrosion by the method of galvanization.



28. A student took four test tubes naming A, B, C and D. He filled each with following solutions:
A- Copper sulphate solution
B- Iron sulphate solution
C- Zinc sulphate solution
D- Silver nitrate solution

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He then put a small iron nail in each test tube. After sometime he finds colour change in some test tubes.

(a) Which test tube will show change in colour and why? (2)
(b) Write the chemical reaction involved. (1)
(c) What type of reaction is this? (1)

29. Attempt either option A or B

A.

(a) 'A' and 'B' are sodium salts of long chain carboxylic acid and long chain sulphonic acid respectively. Which one of 'A' or 'B' will you prefer as a cleansing agent while using underground water (hand pump water)? Give the reason for your answer. (2)

(b) Elaborate on the process of cleansing action. Illustrate micelle with the help of a labelled diagram. (2)

(c) Write the chemical equation of the preparation of soap from an Ester CH_3COOCH_3. What is the name of this process? (1)

OR

B.

(I) Write the next homologue of each of the following: (1)
(i) C_2H_4
(ii) C_4H_6

(II) Name the following compounds : (1)
(a) $CH_3 - CH_2 - OH$
(b) $CH_3 - \overset{\overset{H}{|}}{C} = O$

(III) Write the name and structure of an alcohol with three carbon atoms in its molecule. (1)

(IV) Which element exhibits the property of catenation to maximum extent and why? (2)

SECTION -C

30. If a wire has resistance R, then the resistance of another wire of same material with double length and same area will be: (1)
(a) R
(b) R/2
(c) 4R
(d) 2R

31. The strength of a magnetic field inside a long current carrying Straight solenoid is (1)
(a) More at the ends than at the centre
(b) Minimum in the middle
(c) Same at all points
(d) Found to increase from one end to the other.

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32. Assertion (A): Alloys are commonly used in electrical heating devices like electric iron toaster etc.

Reason(R): Alloys do not oxidize or burn readily at high temperatures. (1)

33. Draw magnetic field lines around a bar magnet. (2)

34. A person needs a lens of power -4.5 D for correction of his vision.

(a) What kind of defect in vision is he suffering from? (1)

(b) Find out the focal length of the corrective lens. (1)

35. What is solenoid? Draw a diagram to show the magnetic field lines produced around a current carrying Solenoid. List any two factors on which the strength of magnetic field lines in a current carrying solenoid depends. (3)

36. If the image formed by a mirror for all positions of the object placed in front of it is always erect and diminished, what type of mirror is it ? Draw a ray diagram to justify your answer. Where and why do we generally use this type of mirror? (3)

37. A concave lens of focal length 15 cm forms an image at a distance of 10 cm from the lens. How far is the object placed from the lens. Draw the ray diagram to justify your answer . (3)

38. Riya is performing an experiment in her school laboratory using a **concave mirror**. She places an object at different distances from the mirror and observes the image formed on a screen. When the object is placed beyond the centre of curvature, she notices that the image is formed between the focus and the centre of curvature. The image is real, inverted, and smaller in size.

Later, Riya uses a glass slab and observes that when a ray of light passes through it obliquely, it bends at the air–glass interface and again at the glass–air interface. However, the emergent ray remains parallel to the incident ray.

Based on the above observations, answer the following questions: (4)

(a) What happens to the speed of light when it enters a glass slab from air

(b) Is the image formed by a glass slab real or virtual

(c) Why does the emergent ray remain parallel to the incident ray in a rectangular glass slab?

OR

What happens to a ray of light when it passes through a rectangular glass slab placed obliquely?

39. (a) Why is the series arrangement not used for domestic circuits? (1)

(b) An electric heater of resistance 8 ohm draw 15 A from the service mains in two hours . Calculate the rate at which heat is developed in the heater. (2)

(c) Determine the total resistance of the parallel combination of three resistances of 2 ohm, 4 ohm and 6 ohm. (1)

(d) State Ohm's law (1)

OR

(a) A copper wire has diameter 0.5 mm and resistivity 1.6 × 10-8 Ω m. Calculate the length of this wire to make it resistance 100 Ω. How much does the resistance change if the diameter is doubled without changing its length? (3)

(b) Define resistance of a conductor. State the factors on which resistance of a conductor depends. Name the device which is often used to change the resistance without changing the voltage source in an electric circuit. (2)

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